SCIENCE EDUCATION

PHYSICS

A Comprehensive Guide for Class XI

Premium Edition • CBSE Curriculum

CHAPTER 4

Laws of Motion

Understanding Force, Inertia, and the Principles Governing All Motion

In kinematics (Chapters 2 and 3), we described *how* objects move — their position, velocity, and acceleration — without asking *why* they move. Now we turn to **dynamics**, the branch of mechanics that explains the causes of motion. The answer lies in **forces**.

Isaac Newton (1642–1727) formulated three elegant laws that connect force and motion. These laws — the foundation of classical mechanics — explain everything from why you lurch forward when a bus brakes, to how rockets propel themselves in the vacuum of space, to the orbits of planets. They remained unchallenged for over 200 years until Einstein's relativity modified them for objects near the speed of light.

This chapter introduces Newton's three laws, the concept of momentum, common forces in nature (friction, tension, normal force), and problem-solving techniques using free-body diagrams. Mastering these ideas is essential for all subsequent chapters in mechanics.

📜 **Historical Background: From Aristotle to Newton — 2000 Years to Understand Force**

Aristotle (384–322 BC) dominated thinking about force for nearly two millennia. He believed force was needed to *maintain* motion, that heavier objects fall faster, and that the "natural state" of all earthly objects is rest. His physics was built on philosophical reasoning, not experiments.

Galileo Galilei (1564–1642) shattered Aristotle's framework through careful experiments with inclined planes and pendulums. He demonstrated that objects naturally maintain their velocity (inertia) and that force is needed only to *change* motion, not sustain it.

René Descartes (1596–1650) introduced the concept of "quantity of motion" (mass × velocity) — an early version of momentum — and proposed that the total quantity of motion in the universe is conserved.

Isaac Newton (1642–1727) synthesized the work of Galileo, Descartes, Kepler, and others into his three laws of motion, published in *Philosophiæ Naturalis Principia Mathematica* (1687) — widely considered the most important scientific book ever written. Newton wrote the Principia in just 18 months, reportedly motivated by a question from **Edmond Halley** about planetary orbits. The Principia unified terrestrial mechanics (falling apples) with celestial mechanics (orbiting planets) for the first time in human history.

4.1 The Concept of Force

📘 **Definition**

Force: A push or pull that can change the state of rest or motion of an object, change its speed, change its direction of motion, or change its shape. Force is a *vector* quantity — it has both magnitude and direction. SI unit: **newton (N)**. $1\ \text{N} = 1\ \text{kg m s}^{-2}$.

4.1.1 Contact and Non-Contact Forces

Contact Forces	Non-Contact (Field) Forces
Require physical contact between objects	Act at a distance through a field
Friction	Gravitational force
Normal force	Electrostatic force
Tension (in strings/ropes)	Magnetic force
Spring force	Nuclear force
Air resistance / Drag	—
Applied force (push/pull)	—

💡 Everyday Examples

- **Contact:** You push a door (applied force), a book rests on a table (normal force), brakes slow a car (friction), a climber hangs by a rope (tension)
- **Non-contact:** An apple falls from a tree (gravity), a comb attracts paper bits after rubbing (electrostatic), a compass needle turns (magnetic)

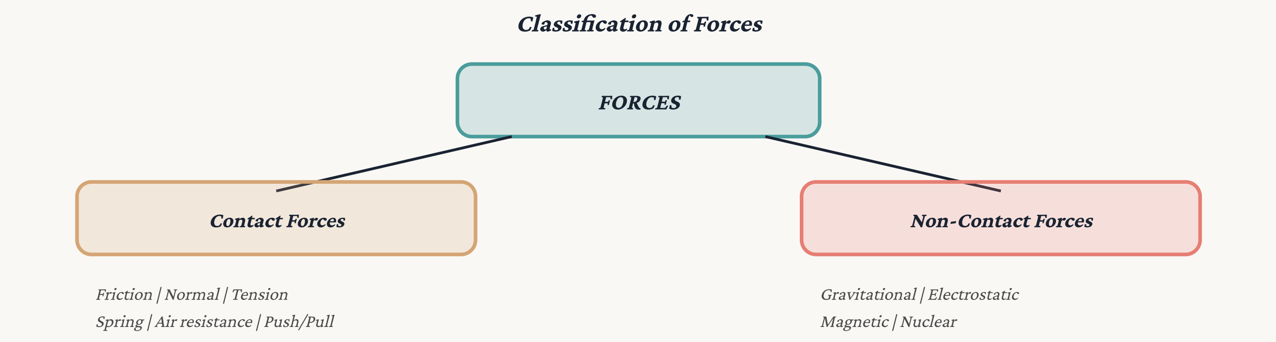

Figure 4.1: Classification of forces into contact and non-contact types

4.2 Historical Background: Aristotle vs Galileo

4.2.1 Aristotle's View (Incorrect)

Aristotle (384–322 BC) believed that a force is needed to keep an object in motion. He thought that if no force acts, the object naturally comes to rest. This seems intuitive — a ball rolling on the ground does slow down and stop. But Aristotle failed to account for **friction**, which is the unseen force causing the ball to stop.

4.2.2 Galileo's Insight (Correct)

Galileo Galilei (1564–1642) challenged this with his famous thought experiment on inclined planes:

Galileo's Inclined Plane Experiment

→ A ball rolling down one inclined plane rises to the same height on an opposite plane (regardless of slope)

→ As the second plane is made less steep, the ball travels further to reach the same height

→ If the second plane is **horizontal** (perfectly smooth), the ball would travel **forever** — it never reaches the original height

→ **Conclusion:** No force is needed to maintain motion; force is needed to *change* motion. An object in motion continues moving at constant velocity unless a force acts on it

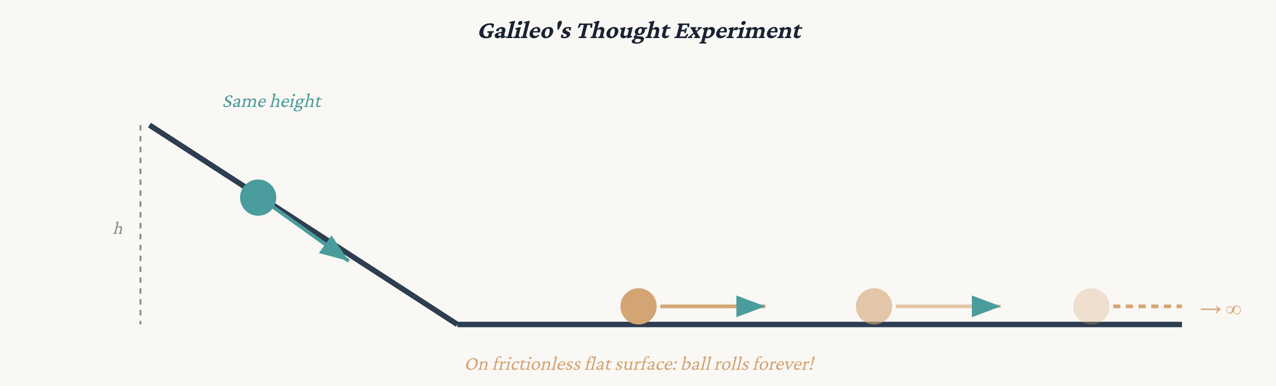

Figure 4.2: Galileo's thought experiment — on a frictionless surface, the ball rolls forever

Aristotle's View	Galileo's View (Correct)
Force is needed to maintain motion	Force is needed to *change* motion
Natural state of objects is rest	Natural state is constant velocity (including rest)
Objects stop because there is no force	Objects stop because of friction (an opposing force)
Held for nearly 2000 years	Led directly to Newton's First Law

4.3 Newton's Laws of Motion

4.3.1 Newton's First Law — The Law of Inertia

📘 **Newton's First Law**

Every object continues in its state of rest or of uniform motion in a straight line unless compelled to change that state by an external unbalanced force.

📘 **Inertia**

Inertia is the inherent property of a body to resist any change in its state of rest or uniform motion. The **mass** of an object is a quantitative measure of its inertia — greater mass means greater inertia.

Types of Inertia (with examples)

→ **Inertia of rest:** Tendency to remain at rest. Example: passengers jerk backward when a bus starts suddenly
→ **Inertia of motion:** Tendency to keep moving. Example: passengers jerk forward when a bus brakes suddenly
→ **Inertia of direction:** Tendency to maintain direction. Example: sparks fly off tangentially from a grinding wheel

🔬

Isaac Newton — Architect of Classical Mechanics

1642–1727, England

Born on Christmas Day 1642 in Woolsthorpe, England, Newton was a premature baby not expected to survive. During the "Annus Mirabilis" (miraculous year) of 1665–66, while Cambridge was closed due to plague, the 23-year-old Newton — working alone at his family farm — developed calculus, the theory of colours, and the law of gravitation. His *Principia* (1687) laid the foundation for all of classical mechanics and is widely regarded as the most important scientific work ever published. He also served as Warden of the Royal Mint and President of the Royal Society. His famous words: *"If I have seen further, it is by standing on the shoulders of giants."*

💡 **Real-Life Examples of First Law**

- A coin on a cardboard falls into a glass when the cardboard is flicked (inertia of rest)
- Dust comes off a carpet when beaten (inertia of rest)
- An athlete continues to run past the finish line (inertia of motion)
- Seat belts prevent you from flying forward in a sudden stop (inertia of motion)

The First Law also defines the concept of an **inertial frame of reference** — a frame in which the first law holds true. Any frame moving with constant velocity relative to an inertial frame is also inertial. Accelerating frames are non-inertial.

4.3.2 Newton's Second Law — F = ma

📘 **Newton's Second Law**

The rate of change of momentum of a body is directly proportional to the applied force and takes place in the direction of the force.

Mathematically: $\vec{F} = d\vec{p}/dt$
For constant mass: $\vec{F} = m\vec{a}$ (Force = mass × acceleration)

NEWTON'S SECOND LAW

$$\vec{F}_{net} = m\vec{a} \quad \text{(for constant mass)}$$

$$\vec{F} = d\vec{p}/dt = d(m\vec{v})/dt \quad \text{(general form)}$$

$$\text{In component form: } F_x = ma_x, \quad F_y = ma_y$$

Key Points about the Second Law

→ $\vec{F}$ here is the **net (resultant) external force**, not any single force
→ Acceleration is in the **same direction** as the net force
→ The First Law is a special case: when $F_{net} = 0$, $a = 0$, so v = constant
→ 1 newton = force needed to give a 1 kg mass an acceleration of 1 m/s²
→ If multiple forces act: $\vec{F}_{net} = \vec{F}_1 + \vec{F}_2 + \vec{F}_3 + \ldots = m\vec{a}$

💡 Solved Example 4.1

Q: A force of 20 N acts on a body of mass 4 kg at rest. Find (a) acceleration, (b) velocity after 5 s, (c) distance in 5 s.

Solution:

(a) $a = F/m = 20/4 =$ **5 m/s²**
(b) $v = u + at = 0 + 5(5) =$ **25 m/s**
(c) $s = ut + \frac{1}{2}at^2 = 0 + \frac{1}{2}(5)(25) =$ **62.5 m**

📜 How Newton Actually Stated the Second Law

Newton did NOT originally write "F = ma." His actual statement in the *Principia* was: *"The change of motion is proportional to the motive force impressed, and is made in the direction of the straight line in which that force is impressed."* Here, "change of motion" means change in momentum (Δp). The familiar form F = ma was developed later by **Leonhard Euler (1707–1783)**, the great Swiss mathematician who reformulated Newton's laws into the differential equation form we use today. Many results attributed to "Newtonian mechanics" were actually formalized by Euler.

4.3.3 Newton's Third Law — Action and Reaction

📘 Newton's Third Law

For every action, there is an equal and opposite reaction. When object A exerts a force on object B, then object B simultaneously exerts a force on object A that is equal in magnitude and opposite in direction.

$$\vec{F}_{AB} = -\vec{F}_{BA}$$

Crucial Points about the Third Law

→ Action and reaction act on **different bodies** — they never cancel each other
→ They are **simultaneous** — neither comes first
→ They are always **equal in magnitude** and **opposite in direction**
→ They are of the **same type** (both gravitational, both contact, etc.)

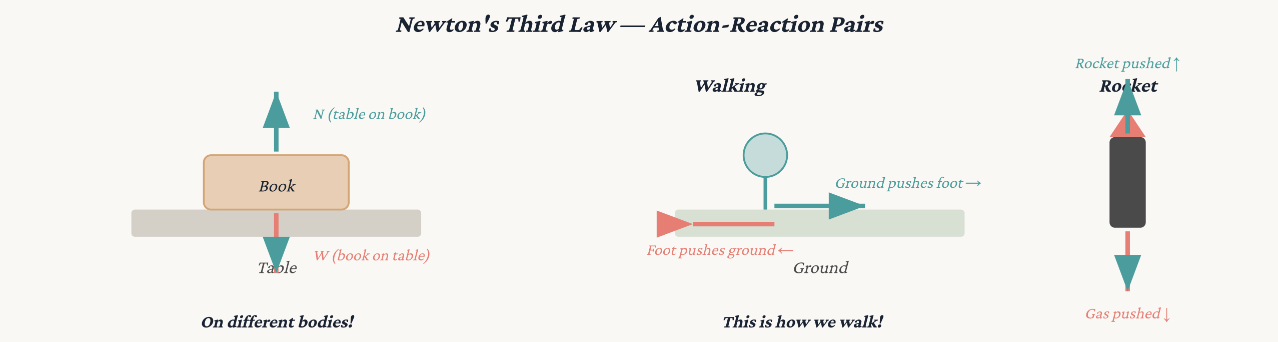

Figure 4.3: Action-reaction pairs — book/table, walking, and rocket propulsion

⚠ **Common Misconception**

"If action = reaction, how does anything move?"

Action and reaction act on *different* bodies! When you push a wall, the wall pushes you back (reaction). But the forces are on different objects. The net force on *you* depends on all forces acting on you — the wall's push, friction on your feet, gravity, etc. Motion happens when the net force on an object is non-zero.

4.4 Common Types of Forces

4.4.1 Friction

📘 **Definition**

Friction: A contact force that opposes the relative motion (or attempted motion) between two surfaces in contact. It acts *parallel* to the contact surface and *opposite* to the direction of motion or tendency of motion.

📜 **The Science of Friction — da Vinci to Coulomb**

Leonardo da Vinci (1452–1519) was the first to systematically study friction around 1493. He discovered two key laws: (1) friction is proportional to the load (normal force), and (2) friction is independent of the contact area. However, his notes were unpublished for centuries.

Guillaume Amontons (1663–1705) independently rediscovered da Vinci's laws in 1699 and published them — these are sometimes called "Amontons' laws."

Charles-Augustin de Coulomb (1736–1806) — famous for Coulomb's law in electricity — distinguished between static and kinetic friction in 1785 and showed that kinetic friction is approximately independent of velocity. The modern friction equations ($f = \mu N$) bear the contributions of all three scientists.

Static Friction (f_s)

Friction that acts when the object is **not moving** but a force is trying to move it. It adjusts its value to match the applied force, up to a maximum limit.

STATIC FRICTION

$$f_s \leq \mu_s N \quad \text{(self-adjusting, up to maximum)}$$

$$\text{Maximum static friction: } f_{s(max)} = \mu_s N$$

$$\text{where } \mu_s = \text{coefficient of static friction, } N = \text{normal force}$$

Kinetic (Sliding) Friction (f_k)

Friction that acts when the object is **already moving** (sliding) over the surface. It has a constant value for a given pair of surfaces.

KINETIC FRICTION

$$f_k = \mu_k N \quad \text{(constant value)}$$

Always: $\mu_k < \mu_s$ (kinetic friction < max static friction)

This is why it's harder to START moving an object than to KEEP it moving

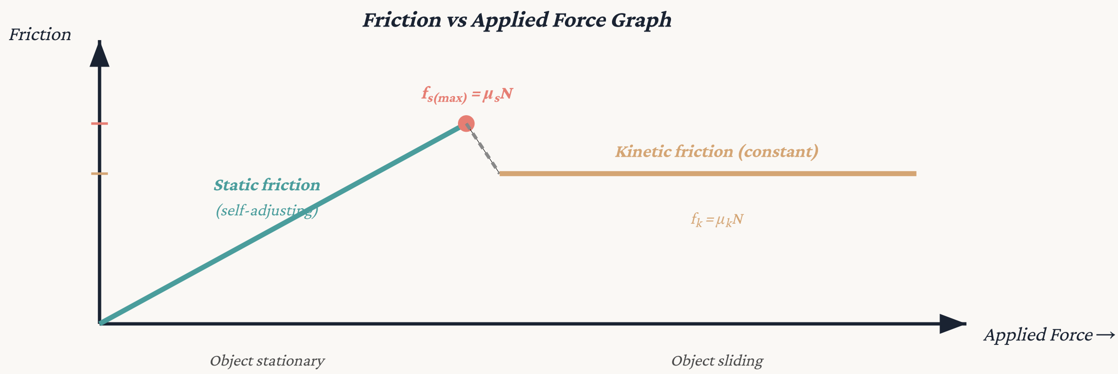

Figure 4.4: Friction increases with applied force up to $f_{s(max)}$, then drops to constant f_k

4.4.2 Normal Force (N)

📘 **Definition**

Normal Force: The contact force exerted by a surface on an object resting on it, acting *perpendicular* (normal) to the surface. It prevents objects from passing through surfaces. On a horizontal surface with no other vertical forces: **N = mg**.

4.4.3 Tension (T)

📘 **Definition**

Tension: The pulling force transmitted through a string, rope, cable, or chain when it is pulled tight by forces at both ends. For a *massless, inextensible* string, the tension is the same throughout its length. Tension always *pulls* (never pushes).

💡 **Solved Example 4.2**

Q: A block of mass 10 kg rests on a horizontal floor. $\mu_s = 0.4$, $\mu_k = 0.3$, $g = 10$ m/s². Find (a) maximum static friction, (b) kinetic friction, (c) acceleration if a 50 N horizontal force is applied.

Solution: N = mg = 10 × 10 = 100 N

(a) $f_{s(max)} = \mu_s N = 0.4 \times 100 =$ **40 N**

(b) $f_k = \mu_k N = 0.3 \times 100 =$ **30 N**

(c) Since applied force (50 N) > $f_{s(max)}$ (40 N), the block moves.

Net force = $F - f_k = 50 - 30 = 20$ N

$a = F_{net}/m = 20/10 =$ **2 m/s²**

4.5 Momentum

4.5.1 Linear Momentum

Definition

Linear Momentum ($\vec{p}$): The product of an object's mass and velocity. It is a *vector* quantity in the direction of velocity.

$\vec{p} = m\vec{v}$
SI unit: **kg m s⁻¹** (or N s). Dimensions: $[M L T^{-1}]$

René Descartes — Pioneer of the Momentum Concept

1596–1650, France

Before Newton, it was Descartes who first proposed that the "quantity of motion" (mass × speed) is conserved in the universe — a profound insight, though his formulation lacked the directional (vector) character of true momentum. Newton refined this into the vector quantity $\vec{p} = m\vec{v}$ and built his entire Second Law around it: $F = dp/dt$. Descartes is also famous for inventing the Cartesian coordinate system (the x-y axes used in all our graphs), without which describing motion mathematically would be far more difficult.

Newton's Second Law in terms of momentum: $\vec{F} = d\vec{p}/dt$. This is the more general form — it applies even when mass changes (e.g., a rocket losing fuel).

IMPULSE-MOMENTUM THEOREM

Impulse = Change in momentum

$$\vec{J} = \vec{F} \times \Delta t = \Delta\vec{p} = m\vec{v} - m\vec{u}$$

SI unit of impulse: $N\ s = kg\ m\ s^{-1}$

Solved Example 4.3

Q: A ball of mass 0.5 kg moving at 10 m/s is brought to rest in 0.1 s. Find the force applied.
Solution:
Change in momentum = $mv - mu = 0.5(0) - 0.5(10) = -5$ kg m/s
$F = \Delta p/\Delta t = -5/0.1 =$ **−50 N** (negative = opposing direction of motion)
Magnitude of force = **50 N**

4.5.2 Conservation of Linear Momentum

Law of Conservation of Momentum

If no external force acts on a system of particles, the total momentum of the system remains constant.

For a two-body system: $\mathbf{m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2}$
(Total momentum before = Total momentum after)

Proof: From Newton's Third Law, for two interacting bodies: $\vec{F}_{12} = -\vec{F}_{21}$. By Second Law: $d\vec{p}_1/dt = -d\vec{p}_2/dt \rightarrow d(\vec{p}_1 + \vec{p}_2)/dt = 0 \rightarrow \vec{p}_1 + \vec{p}_2 =$ constant.

💡 Solved Example 4.4

Q: A gun of mass 5 kg fires a bullet of mass 50 g at 200 m/s. Find the recoil velocity of the gun.

Solution: Initially both at rest → total initial momentum = 0.

By conservation: $m_1v_1 + m_2v_2 = 0$

$5 \times V + 0.05 \times 200 = 0$

$5V = -10$

V = **−2 m/s** (negative = backward direction)

The gun recoils at 2 m/s in the opposite direction to the bullet.

💡 Solved Example 4.5

Q: A 60 kg person jumps from a boat of mass 200 kg at 4 m/s. Find the boat's velocity.

Solution: Initial momentum = 0 (both at rest).

$60 \times 4 + 200 \times V = 0 \rightarrow V = -240/200 =$ **−1.2 m/s** (opposite direction)

4.6 Equilibrium of a Body

📘 Definition

Equilibrium: A body is in equilibrium when the net external force acting on it is **zero**. The body either remains at rest (*static equilibrium*) or moves with constant velocity (*dynamic equilibrium*).

CONDITIONS FOR EQUILIBRIUM (TRANSLATIONAL)

$$\Sigma\vec{F} = 0 \quad \text{(vector sum of all forces = zero)}$$

$$\text{In component form: } \Sigma F_x = 0 \quad \text{and} \quad \Sigma F_y = 0$$

💡 Solved Example 4.6

Q: A book of mass 2 kg lies on a table. Identify all forces and show equilibrium.

Solution:

Forces on the book: (i) Weight $W = mg = 2 \times 10 = 20$ N (downward), (ii) Normal force N (upward)

For equilibrium: $\Sigma F_y = N - W = 0 \rightarrow N = W =$ **20 N**

The book is in static equilibrium — net force = 0, acceleration = 0.

💡 Solved Example 4.7

Q: Three concurrent forces of 10 N, 10 N, and 10√2 N are in equilibrium. Find the angles between them.

Solution: If two 10 N forces are perpendicular, their resultant = $\sqrt{(100+100)} = 10\sqrt{2}$ N. For equilibrium, the third force ($10\sqrt{2}$ N) must be equal and opposite to this resultant.

So the angle between the two 10 N forces = **90°**, and the $10\sqrt{2}$ N force makes **135°** with each of them.

4.7 Force in Circular Motion — Centripetal Force

From Chapter 3, we know that an object in uniform circular motion has centripetal acceleration $a_c = v^2/r$ directed toward the centre. By Newton's Second Law, this acceleration requires a **net force toward the centre** — the centripetal force.

CENTRIPETAL FORCE

$$F_c = ma_c = mv^2/r = m\omega^2 r$$

Direction: Always toward the centre of the circular path

This is NOT a new force — it is the net radial force provided by existing forces (tension, gravity, friction, normal force, etc.)

Sources of Centripetal Force (Examples)

→ **Stone on a string:** Tension provides centripetal force

→ **Car on a curved road:** Friction provides centripetal force

→ **Planet orbiting Sun:** Gravitational force provides centripetal force

→ **Electron orbiting nucleus:** Electrostatic force provides centripetal force

→ **Banked road:** Component of normal force provides centripetal force

💡 **Solved Example 4.8**

Q: A car of mass 1000 kg moves at 20 m/s on a curved road of radius 200 m. Find the centripetal force needed. What provides this force?

Solution:

$F_c = mv^2/r = 1000 \times 400/200 = \mathbf{2000\ N}$

This force is provided by **friction** between the tires and the road.

💡 Solved Example 4.9

Q: What is the maximum speed at which a car can safely negotiate a curve of radius 100 m if μ = 0.5? (g = 10 m/s²)

Solution: Friction = centripetal force (at maximum speed)

$\mu mg = mv^2/r \rightarrow v^2 = \mu gr = 0.5 \times 10 \times 100 = 500$

$v = \sqrt{500} \approx \mathbf{22.4\ m/s} \approx \mathbf{80.5\ km/h}$

4.7.1 Banking of Roads

On a banked road (road tilted at angle θ), the normal force has a horizontal component that provides centripetal force, reducing or eliminating the need for friction.

BANKING ANGLE (NO FRICTION)

$$\tan\theta = v^2/(rg)$$

MAXIMUM SPEED ON BANKED ROAD (WITH FRICTION Μ)

$$v_{max} = \sqrt{[rg(\mu + \tan\theta)/(1 - \mu\tan\theta)]}$$

Why bank roads? On a flat road, only friction provides centripetal force — on wet/icy roads, friction is low and cars skid. By banking (tilting) the road, the normal force's horizontal component provides centripetal force even without friction. This is why highway curves, railway tracks, cycling tracks, and airplane turns are all banked.

💡 **Solved Example 4.12**

Q: A road curve has radius 100 m. Find the banking angle for a design speed of 50 km/h (no friction). (g = 10 m/s²)

Solution: $v = 50 \times 5/18 = 13.9$ m/s

$\tan\theta = v^2/(rg) = (13.9)^2/(100 \times 10) = 193.2/1000 = 0.193$

$\theta = \tan^{-1}(0.193) \approx \mathbf{10.9°}$

4.8 Problem-Solving Techniques

4.8.1 Free-Body Diagrams (FBDs)

📘 **Definition**

Free-Body Diagram (FBD): A diagram showing a single object isolated from its surroundings, with ALL forces acting on it drawn as arrows from the object. This is the most important tool for solving force problems.

Steps for Solving Newton's Law Problems

→ **Step 1:** Identify the object(s) of interest

→ **Step 2:** Draw the Free-Body Diagram — show ALL forces acting on the object (weight, normal, friction, tension, applied force, etc.)

→ **Step 3:** Choose a convenient coordinate system (usually x along motion, y perpendicular)

→ **Step 4:** Resolve forces into components along chosen axes

→ **Step 5:** Apply Newton's Second Law: $\Sigma F_x = ma_x$ and $\Sigma F_y = ma_y$

→ **Step 6:** Solve the resulting equations for unknowns

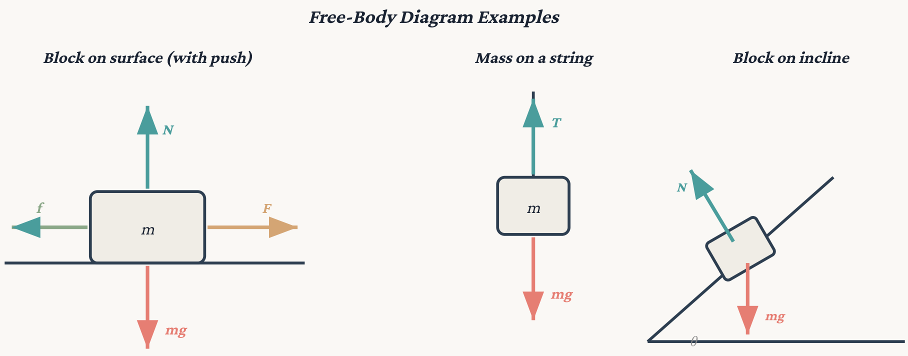

Figure 4.5: Free-body diagrams for common scenarios

💡 Solved Example 4.10 — Connected Bodies (Atwood Machine)

Q: Two masses m_1 = 8 kg and m_2 = 6 kg are connected by a string over a frictionless pulley. Find the acceleration and tension. ($g = 10\ m/s^2$)

Solution: The heavier mass moves down, lighter moves up.

For m_1 (moving down): $m_1g - T = m_1a \rightarrow 80 - T = 8a$ ...(i)

For m_2 (moving up): $T - m_2g = m_2a \rightarrow T - 60 = 6a$ ...(ii)

Adding (i) and (ii): $80 - 60 = (8 + 6)a \rightarrow 20 = 14a$

$a = 20/14 \approx 1.43\ m/s^2$

From (ii): $T = 60 + 6(1.43) = 60 + 8.57 \approx$ **68.6 N**

General formulas: $a = (m_1 - m_2)g/(m_1 + m_2)$, $T = 2m_1m_2g/(m_1 + m_2)$

💡 Solved Example 4.11 — Block on Incline

Q: A block of mass 5 kg is placed on a smooth incline of angle 30°. Find its acceleration down the incline and the normal force. ($g = 10\ m/s^2$)

Solution: Choose axes: x along incline (down = +), y perpendicular to incline (outward = +).

Forces: Weight mg = 50 N, Normal N (perpendicular to surface).

Resolving mg: Along incline = mg sin 30° = 25 N; Perpendicular = mg cos 30° = 43.3 N

Along incline: $ma = mg \sin\theta \rightarrow a = g \sin 30° = 10 \times 0.5 =$ **$5\ m/s^2$**

Perpendicular: $N = mg \cos 30° = 50 \times (\sqrt{3}/2) \approx$ **43.3 N**

Note: On an incline, $N \neq mg$! $N = mg\cos\theta < mg$.

📌 Chapter Summary — Must Remember

- Force: push or pull; vector; SI unit = newton (N); contact vs non-contact
- Galileo: force changes motion (not maintains it); Aristotle was wrong
- **1st Law:** Object remains at rest or in uniform motion unless acted upon by external force; defines inertia
- **2nd Law:** $\vec{F} = m\vec{a} = d\vec{p}/dt$; acceleration is in the direction of net force
- **3rd Law:** Action = −Reaction; they act on DIFFERENT bodies
- Friction: $f_s \leq \mu_s N$ (static); $f_k = \mu_k N$ (kinetic); $\mu_k < \mu_s$
- Normal force is perpendicular to surface; on incline $N = mg\cos\theta$

- Momentum: $\vec{p} = m\vec{v}$; impulse $J = F\Delta t = \Delta p$
- Conservation of momentum: when $F_{ext} = 0$, total $\vec{p}$ = constant
- Equilibrium: $\Sigma\vec{F} = 0 \rightarrow a = 0$ (rest or constant velocity)
- Centripetal force: $F_c = mv^2/r$ (not a new force — it's the net inward force)
- FBD is the essential first step in all force problems

⚠ **Common Mistakes to Avoid**

- Thinking action and reaction cancel — they act on different bodies!
- Forgetting friction when an object moves on a surface
- Assuming $N = mg$ on inclined surfaces ($N = mg\cos\theta$ on inclines)
- Using kinetic friction when the object hasn't started moving yet
- Forgetting that centripetal force is provided by existing forces, not a separate force
- Not drawing FBDs before solving problems
- Confusing mass and weight (mass in kg, weight in N; $W = mg$)
- Applying conservation of momentum when external forces exist

Practice Questions (Unsolved)

Conceptual Questions

1. If action = reaction, how does any object ever accelerate?
2. Why is it easier to push a moving box than to start it?
3. How does a rocket propel in vacuum?
4. A horse pulls a cart equally backward. How does the system move?
5. Explain how friction can be both helpful and harmful.
6. Why do passengers jerk forward when a bus brakes?
7. Is centripetal force a separate force? What provides it?

Short Answer Questions (2-3 Marks)

1. State Newton's three laws with one example each.
2. Prove conservation of momentum from Newton's third law.
3. Explain static and kinetic friction. Why is $\mu_k < \mu_s$?
4. Draw FBD for a block on an inclined plane and resolve forces.
5. Define impulse. Show impulse = change in momentum.

Numerical Problems

1. 50 N on 10 kg at rest. Find acceleration, velocity after 4 s, distance in 4 s.
2. 10 kg block, $\mu S=0.4$, $\mu K=0.3$, pushed with 50 N. Find friction and acceleration. (g=10)
3. Gun 5 kg fires 50 g bullet at 200 m/s. Find recoil velocity.
4. Atwood machine: $m_1=10$ kg, $m_2=8$ kg. Find a and T. (g=10)
5. 5 kg on smooth 30° incline. Find acceleration and normal force. (g=10)
6. Max speed on curve r=50 m, $\mu=0.4$. (g=10)
7. 0.1 kg ball at 20 m/s hit back at 30 m/s in 0.01 s. Find force.

Assertion-Reason Questions

(a) Both true, R explains A (b) Both true, R not explanation (c) A true, R false (d) A false, R true

1. **A:** Action-reaction never cancel. **R:** They act on different bodies.
2. **A:** Body in equilibrium can be in motion. **R:** Equilibrium requires zero net force, not zero velocity.
3. **A:** On incline, $N = mg\cos\theta$ not mg. **R:** Only perpendicular component is balanced by N.
4. **A:** Friction is self-adjusting up to a limit. **R:** Static friction matches applied force until μSN.